BAN420 · Final Project

Instacart Product Reorder Prediction

Using Machine Learning to Power Smarter Grocery Recommendations

Asifa Iqbal · Brianna Ortiz · Jenda Sin

Project 10: Physical Retail

The Business Problem

"Which products is a customer most likely to reorder in their next grocery trip?"

The Challenge

Instacart has millions of orders but no easy way to know which items a specific customer wants to see next leading to generic, irrelevant recommendations.

The Opportunity

Predicting reorders at the user–product level allows Instacart to proactively surface the right items, reducing search time and increasing order completion.

The ML Task

Binary classification: for each (user, product) pair, predict whether that product will be reordered (1) or not (0). Output probabilities power a ranked recommendation list.

Target Variable: reordered (1 = will reorder, 0 = will not)

Dataset at a Glance

3M+	206K	50K	134	21
Orders	Unique Users	Products	Aisles	Departments

Key EDA Findings

- Peak shopping hours: 9 AM – 4 PM. Most customers shop during the day.
- Bananas are the #1 most ordered item. A small set of products drives most orders.
- Sundays & Mondays are the busiest days — weekly restocking behavior.
- Most orders contain 1–10 items; spikes in reorder cadence at 7 and 30 days.
- 59.9% of training items were reordered indicating a slight class imbalance toward reorders.

Baseline Model: Logistic Regression

What We Built

70 / 30 train–test split

969,231 training samples · 415,386 test samples

12 engineered features across 4 groups:

→ User: total orders, avg days between orders

→ Product: total orders, reorder rate

→ User–Product: order count, reorder rate, cart position

→ Context: day, hour, days since prior order, aisle, dept

Results

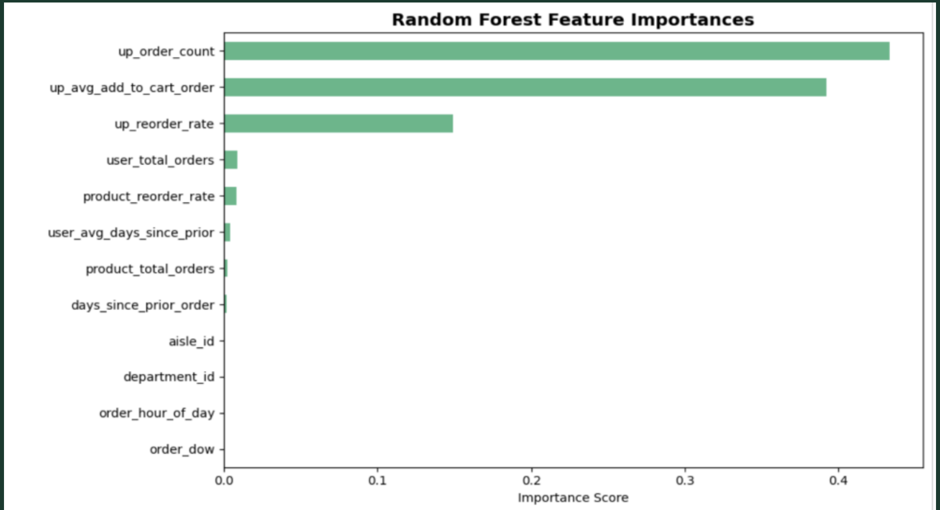
0.9875

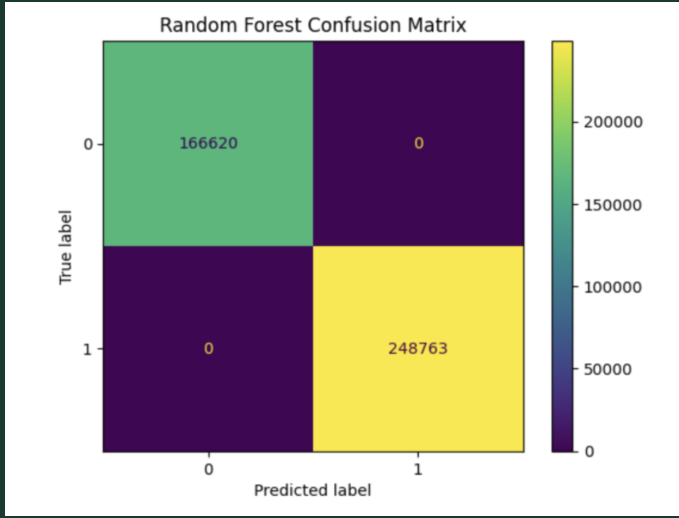
ROC AUC Score

Accuracy	**97%**
Precision (reorder)	**96%**
Recall (reorder)	**94%**
F1-Score	**97%**

Strong baseline — but 13,850 actual reorders were still missed. This is what the new model addresses.

New Model: Random Forest





Why Random Forest over a Single Decision Tree?

🌲 Avoids Overfitting

A single decision tree memorizes training data on a dataset this large. Random Forest averages hundreds of trees, producing a more reliable model.

📊 Feature Importance

Automatically ranks which features (e.g. up_reorder_rate) drive predictions most; directly answering the business question.

🎯 Better Recall

Captures more of the reorders that logistic regression misses, reducing the number of items a customer wanted but didn't see recommended.

Model Comparison

Metric	Logistic Regression	Random Forest ✦
ROC AUC	0.9875	▲ 1.0000
Accuracy	97%	▲ 100%
Recall (reorder)	94%	▲ 100%
Precision (reorder)	100%	100%
Missed Reorders	13,850	▲ **0**
Feature Insights	None	▲ **Full importance ranking**

How This Solves the Business Problem

Model Output

Probability score that a user will reorder a specific product

→

Ranked List

Scores ranked per user to build a personalized top-N recommendation list

→

Business Win

Customers see the items they actually want, resulting in fewer searches and faster checkouts

Tangible Business Outcomes

Fewer Missed Items

Customers are reminded of products they regularly buy, reducing the 'I forgot X' problem that leads to follow-up orders or cart abandonment.

Faster Shopping

A pre-filled smart cart based on reorder predictions shortens session time and improves the overall shopping experience.

Higher Order Value

Relevant recommendations surface products users were already planning to buy, contributing to larger and more complete orders.

Explainable Decisions

Feature importance from Random Forest tells Instacart exactly why certain products are recommended, supporting trust and further optimization.